PEDESTRIAN MODELING AND TRAJECTORY PREDICTION FOR SOCIALLY COMPLIANT NAVIGATION IN ROBOTIC SYSTEM

A THESIS

submitted by

PRERANA KUMARI

(*Roll No.*: **M241109EE**)

in partial fulfillment of the requirements for the award of the degree of

Master of Technology

in

Electrical Engineering

(Industrial Power and Automation)

Under the guidance of

Dr. Ranjith Ravindranathan Nair

Department of Electrical Engineering

NATIONAL INSTITUTE OF TECHNOLOGY CALICUT

NIT Campus P.O., Kozhikode - 673601, India

JUNE 2026

CERTIFICATE

This is to certify that the thesis entitled "**Pedestrian Modeling And Trajectory Prediction For Socially Complaint Navigation In Robotic System**" is a bonafide record of the **project work** carried out by **Prerana Kumari** (Roll No.: **M241109EE**) under my supervision, in partial fulfillment of the requirements for the award of the degree of **Master of Technology** in **Electrical Engineering (Industrial Power and Automation)** at **National Institute of Technology Calicut**. This work has not been submitted elsewhere for the award of any degree or diploma.

Dr. Ranjith Ravindranathan Nair
(Guide)
Assistant Professor
Department of Electrical Engineering

Dr.Jagadanand G.
Professor & Head
Department of Electrical Engineering

Place: NIT Calicut
Date:

DECLARATION

I, **Prerana Kumari** (Roll No.: **M241109EE**) hereby declare that this project report entitled **"Pedestrian Modeling And Trajectory Prediction For Socially Complaint Navigation In Robotic System"** is a record of the original work done by me under the guidance of **Dr. Ranjith Ravindranathan Nair**, Assistant Professor, Department of Electrical Engineering, National Institute of Technology Calicut and that this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or any other similar title or recognition to any candidate of any Institute or University, to the best of my knowledge.

Prerana Kumari

(M241109EE)

Place: NIT Calicut

ACKNOWLEDGEMENT

The successful completion of any task is incomplete and meaningless without giving any due credit to the people who made it possible without which the project would not have been successful and would have existed in theory. First and foremost, I owe a lot of thanks to my guide, **Dr Ranjith Ravindranathan Nair**, Asst. Professor, Department of **Electrical Engineering** for constantly motivating me and guiding me through the idea of a creatively and amazingly performed Project in undertaking this endeavor and challenge and also for being there whenever I needed his guidance or assistance also I am thankful to all panel members, **Dr Ranjith Ravindranathan Nair**, **Dr Sunil Kumar T.K.**, **Dr Sandeep J.**, **Dr S Kanagalakshmi** and **Dr Sathiya S**, for their value based addition to correcting at every point to proceed successfully this project.I am grateful to **Dr.Jagadanand G.**, Head of Department of **Electrical Engineering**, and all other faculty members of our department for their constant guidance, support, and motivation.

PRERANA KUMARI

ABSTRACT

Pedestrian modeling and Trajectory prediction by the robot to reach to the goal safely without any disturbance . so it is find that the disturbance is to be find either static or dynamic that we cannot decide but while trajectory it is to be find that as like human being behavior , in road like situation we find obstacle as car , or any other pole or the other human beings then have to make the trajectory accordingly without harming to self as well as the other human well beings , so there also robot has to navigate its path and reach to the goal without disturbing the other human being and harming to the self structure . so that first use machine learning to understand the behavior of human beings the after understand all the cohesion factors that come with the human beings . accordingly understand and make the robot to understand as well as accordingly demonstrate the path .further the same process is performed with the turtle Bot waffle pi and work accordingly to make their trajectory to reach to the goal.This work demonstrate the highlight of social character of the robot.

Contents

List of Abbreviations

APF	Artificial Potential Field
CNN	Convolutional Neural Network
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DeepSORT	Deep Simple Online and Realtime Tracking
LiDAR	Light Detection and Ranging
LSTM	Long Short-Term Memory
ML	Machine Learning
PFM	Potential Field Method
PID	Proportional Integral Derivative Controller
RGB	Red Green Blue
RNN	Recurrent Neural Network
ROS	Robot Operating System
ROS2	Robot Operating System 2
RViz	ROS Visualization Tool
SFM	Social Force Model
SLAM	Simultaneous Localization and Mapping
YOLO	You Only Look Once

List of Figures

Chapter 1

Introduction

Social robots that can perceive their environment, plan, and execute collision-free trajectories are increasingly being deployed in complex human environments. Mobile social robots are becoming more popular in dynamic settings such as museums, airports, shopping malls, and urban areas. These robots are increasingly used in crowded indoor and outdoor environments. Applications include surveillance mainly in airports, healthcare assistance, hospital support, delivery, logistics, and more.
A social robot is an autonomous robot that interacts and communicates with humans or other autonomous physical agents by following social behaviors and rules attached to its role. Like other robots, a social robot is physically embodied (avatars or on-screen synthetic social characters are not embodied and thus distinct). Some synthetic social agents are designed with a screen to represent the head or 'face' to dynamically communicate with users. Social interactions are likely to be cooperative, but the definition is not limited to this situation. Moreover, uncooperative behavior can be considered social in certain situations. The robot could, for example, exhibit competitive behavior within the framework of a game.
The robot could also interact with a minimum or no communication. It could, for example, hand tools to an astronaut working on a space station. However, it is likely that some communication will be necessary at some point.Robots are becoming more common in homes, hospitals, factory floors, and other human environments. Human society operates on mutually accepted social norms, and adhering to these norms is a key indicator of social participation. For robots to be socially compatible with humans,

it is essential for them to follow these social norms.

Due to the inherent difficulty in replicating emotions, any subtle nuances in conversations that robots are not programmed to understand might be perceptible to the human, potentially resulting in interactions between social robots and humans that feel artificial and unrealistic . If social robots do not align with even subtle human social norms, it may decrease the likelihood of humans treating them as equals or perceiving them as having consciousness.

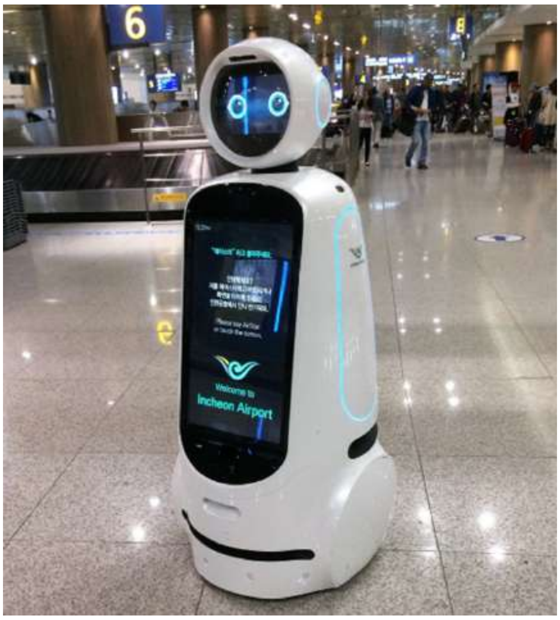

Figure 1.1: Surveillance robot in airport

Fig.1.1 [25] shows the robot used for surveillance purposes outside the country such type of social robots are available . Here in airport Robot is doing surveillance that helps the other to either in sense of security or other day to day activity that is required in small purpose get the help from such type of robot.

1.1 Motivation

Key issues faced by social robots include sudden or unpredictable turns, freezing in the presence of obstacles, and the inability to maintain safe physical distances while navigating.

Ensuring human physical safety involves maintaining an appropriate distance from human beings during robot navigation.
Robots must adhere to social norms by maintaining personal space and minimizing disruption in dynamic human environments.
Furthermore, human psychological safety is a growing concern, as social interactions with robots must not cause stress or discomfort during navigation and communication.

1.2 Background study

With the rapid advancement of autonomous systems, socially aware robot navigation has become an important research area in robotics and artificial intelligence. Mobile robots operating in crowded environments such as shopping malls, airports, hospitals, and university campuses must navigate safely while respecting human social behaviors. Traditional navigation methods focus primarily on obstacle avoidance often neglecting the social interactions that naturally occur among pedestrians. As a result, robots may generate paths that are technically feasible but socially inappropriate, causing discomfort and reducing human acceptance of robotic systems [1, 2].

Pedestrian modeling plays a fundamental role in enabling socially compliant navigation. Pedestrians exhibit complex movement patterns influenced by individual goals, surrounding environments, and interactions with nearby people. Human movement is rarely random; instead, it follows observable social rules such as maintaining personal space, walking in groups, avoiding collisions, and adapting to dynamic changes in the environment. Accurate modeling of these behaviors allows autonomous robots to predict future pedestrian motions and plan safer trajectories. Several studies have emphasized the importance of integrating human-aware navigation strategies into robotic systems to improve safety and social acceptance [3, 4].

One of the most influential approaches for modeling pedestrian behavior is the Social Force Model (SFM), originally proposed by Helbing and Molnar. The model represents pedestrian motion as the result of attractive and repulsive forces acting on individuals. Attractive forces drive pedestrians toward their destinations, while repulsive forces help maintain personal space and avoid collisions with other pedestrians and obstacles.

Due to its simplicity and effectiveness, the Social Force Model has been widely adopted in crowd simulation, pedestrian dynamics analysis, and robot navigation research [5]. Various extensions of the model have been developed to better capture group behavior, environmental influences, and interaction dynamics in crowded scenarios [6].

Pedestrian groups represent a significant aspect of human crowd behavior. People often move together in families, friends, or work groups while maintaining social relationships. Research has shown that group members tend to exhibit coordinated movement patterns characterized by similar walking speeds, close proximity, and frequent visual interactions. These social characteristics affect crowd flow and influence the navigation decisions of nearby pedestrians and robots. Understanding group behavior is therefore essential for developing socially compliant robotic navigation systems [7,8]. Recent studies have introduced cohesion-based metrics to quantify the strength of social relationships within pedestrian groups and analyze their impact on collective movement patterns [9].

Advancements in computer vision and deep learning have significantly improved pedestrian detection and tracking capabilities. Modern object detection algorithms such as YOLO (You Only Look Once) provide real-time detection performance with high accuracy, making them suitable for robotic applications operating in dynamic environments. When combined with multi-object tracking algorithms such as DeepSORT, these systems can continuously track pedestrian identities across video frames and generate reliable trajectory information. Such tracking frameworks serve as the foundation for higher-level behavioral analysis, including velocity estimation, group detection, and trajectory prediction [10, 11].

Trajectory prediction is another critical component of socially aware navigation. Predicting future pedestrian positions enables robots to anticipate potential interactions and make proactive navigation decisions. Traditional trajectory prediction methods rely on kinematic models and linear motion assumptions. However, these approaches often fail in crowded environments where pedestrian movements are strongly influenced by social interactions. To address this limitation, recent research has explored machine learning and deep learning techniques that learn complex motion patterns directly from observed trajectories. Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and attention-based architectures have demonstrated promising performance in predicting pedestrian trajectories in dynamic environments [12, 13].

In addition to trajectory prediction, the concept of group cohesion has gained increasing attention in pedestrian behavior analysis. Cohesion measures the degree of social connectedness among group members and can be evaluated through multiple factors such as spatial proximity, velocity similarity, interaction patterns, and group size. A highly cohesive group typically exhibits coordinated movement, maintains stable interpersonal distances, and responds collectively to environmental changes. These cohesion indicators provide valuable information for understanding social dynamics and can improve trajectory prediction accuracy by incorporating group-level behavioral constraints [7, 9].

Recent developments in socially compliant navigation have focused on integrating pedestrian tracking, group detection, cohesion analysis, and trajectory prediction into a unified robotic framework. Such systems enable robots to understand the social structure of their surroundings navigation strategies that respect human comfort and safety. By considering both individual pedestrian behavior and group dynamics, autonomous robots can achieve more natural interactions with humans while maintaining efficient navigation performance. These advances have opened new opportunities for deploying robotic systems in complex real-world environments where social awareness is essential for successful operation [2, 4, 14].

Therefore, this work focuses on pedestrian modeling and trajectory prediction for socially compliant navigation by combining pedestrian detection, multi-object tracking, group identification, cohesion analysis, velocity estimation, and future trajectory prediction. The integration of these components aims to provide a comprehensive understanding of pedestrian behavior, enabling autonomous robots to navigate safely and efficiently in dynamic human-populated environments.

1.3 Research gap

1. Under normal crowd densities, nearby groups often remain unaffected by the presence of a robot. Existing models frequently treat group cohesion as a lin-

ear combination of features without considering important attributes such as age, gender, and cultural differences.

2. The prediction of future human states, trajectories in dynamic social zones, and estimation of socially acceptable approach poses are often overlooked in existing navigation frameworks.

3. Most current socially aware navigation systems are primarily designed for indoor environments and face challenges when deployed in outdoor or densely populated scenarios.

4. Although social factors are considered during clustering, robots often fail to perform an in-depth analysis of group behavior when clustering is based solely on cluster centers. A better understanding of clustering behavior can improve the robot's ability to approach and interact with humans effectively.

5. Existing approaches heavily depend on the accuracy of pedestrian feature detection. Performance may degrade significantly under challenging conditions such as poor illumination, occlusions, and crowded environments.

1.4 Problem statement

To design and develop socially complaint path planning for socially interference robot using Machine Learning to understand the human Cohesion deployed for surveillance application in human populated environment.

1.5 Objectives

1. To develop a socially aware autonomous navigation system for mobile robots operating in both simulated and real-world environments.

2. To develop a robust control strategy capable of real-time execution in dynamic and human-populated environments.

3. To create pedestrian dynamic models from visual data acquired through cameras and sensing systems.

4. To design and implement a human cohesion analysis algorithm using recorded video data for evaluating pedestrian cohesion factors and assessing their suitability for live video applications.

5. To develop socially intelligent path-planning algorithms using image processing techniques based on proximity grouping, cohesion score estimation, and overall group cohesion analysis from live camera inputs.

6. To design group clustering algorithms that identify pedestrian groups based on cohesiveness, estimate group size, distinguish between small and large groups, and facilitate smooth trajectory generation for robot navigation.

7. To design and implement a YOLO-based pedestrian detection and tracking framework in the Gazebo simulation environment and visualize the results in RViz.

8. To analyze and validate the proposed algorithms on real hardware platforms for overall cohesion estimation in small groups and to generate complete goal-oriented navigation trajectories.

1.6 Organization of the thesis

This thesis includes five chapters, beginning with the introduction, followed by second chapter covers socially complaint path planning their basically shows the work for robots completing the path to reach to the goal. Then third chapter includes the main working area that is pedestrian dynamic modeling which represents to under pedestrian cohesion, its behaviour then robot make the needful trajectory. Next fourth chapter covers the clustering of group of people behaviour to understand their groupism with different clustering method to apply. Last fifth chapter covers the workings of previous chapters simulation to implement in Hardware to make the use of these Machine Learning method to complete the robots trajectory to reach to the Goal.

Chapter 2

Socially complaint path planning

2.1 Introduction

This chapter covers the employment of turtlebot waffle pi in Gazebo and using PFM (Potential field Method), this can be follow to achieve goal oriented trajectory starting with the empty world , then turn towards the Gazebo world should have includes obstacle firstly shows their the obstacle available is of static nature then this obstacle world has to be gone through that robot experimented again in Gazebo by applying APF(Artificial potential field)to make the robot to move towards the goal by avoiding these obstacles coming in the path and reach to the Goal.

2.2 Path planning overview

Socially complaint path planning covers the complete path planning socially to reach to the goal. Here PFM plays a major role to make the trajectory to reach goal.(PFM) is a classical and intuitive approach to robot path planning and obstacle avoidance. In PFM, the robot navigates by following a virtual force field: The goal generates an attractive potential, pulling the robot toward it. Obstacles, including humans or walls, generate repulsive potentials, pushing the robot away to avoid collisions.The robot moves in the direction of the resultant force, calculated as the vector sum of all attractive and repulsive forces acting on it. Now it do apply Leader-Follower Strategy, Actual The Leader-Follower Strategy is a decentralized multi-robot coordination method where one robot (the leader) determines the path, and one or more followers replicate its motion while maintaining a defined spatial relationship (distance and/or formation). Leader: This robot is responsible for navigating the environment, making high-level decisions, avoiding obstacles, and selecting paths (possibly using potential fields or map-based planning). Followers: These robots track the leader's position and orientation, adjusting their own trajectories to maintain a desired formation (e.g., line, triangle, or column) while avoiding collisions with the leader, obstacles, or each other. It works as Position Tracking Followers continuously estimate the position of the leader. This can be done using: Relative position sensors (e.g., LiDAR, camera, ultrasonic sensors), Global coordinates via GPS or SLAM, Visual tracking (e.g., using YOLO + DeepSORT to identify and follow the leader robot) Control strategy:The control strategy defines how the robot translates planning and perception data into motion commands (i.e., linear and angular velocities) that safely and efficiently guide it toward its goal while respecting human presence and maintaining formation in multi-robot settings. Types of Control Strategies Used: 1.Low-Level Motion Control- Controls the robot's actuators (e.g., wheels) to follow desired velocity commands. Typically implemented using PID controllers or built-in ROS controllers like diff-drive-controller. 2.Velocity-Based Control- The robot is controlled through velocity commands. These commands are computed based on: The output of the potential field method Leader's position and orientation Obstacle positions 3.Behavior-Based Control Combines different behavior layers such as: Goal-seeking: move toward the target, Obstacle avoidance: avoid people and static obstacles ,Social compliance: maintain safe distance from humans Formation keeping: maintain position relative to leader

Connect Gazebo to hardware: Simulators mimic the real world, to a certain extent Simulates robots, sensors, and objects in a 3-D dynamic environment Generates realistic sensor feedback and physical interactions between objects.

2.3 Implementation of socially complaint path planning navigation

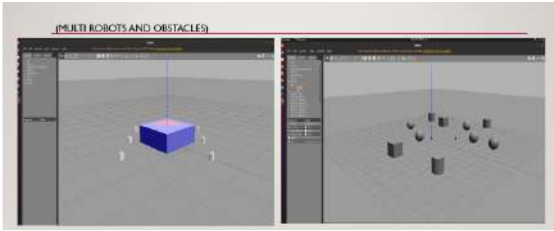

Figure 2.1: Simulation snapshot of multi-robot with obstacle in Gazebo environment

Fig 2.1 Figure shows obstacles like box, ball and multiple turtlebot3 robots in gazebo environment which gives real time like simulation result. These turtle bots have to reach their destination while avoiding obstacles makes collision free intelligent trajectory to reach to the destination.

Figure 2.2: Simulation snapshot of Plotting of the turtlebot trajectory in Gazebo

Fig 2.2 shows obstacles like box, ball and multiple turtlebot3 robots in gazebo environment which gives real time like simulation result. These turtle bots have to reach their destination while avoiding obstacles makes collision free intelligent trajectory to reach to the destination Desired trajectory vs Robot trajectory and the plotting error along with this is shown and its velocity trajectory path is also visualize.

2.4 Summary

Here there is the implementation of Artificial potential method the robot has to be implemented and try to move the robot to make its full trajectory , this trajectory covers the social factor of obstacle while navigating and give the addition of smoothness in trajectory by addition of some controllers to implement in GAZEBO World.To make the trajectory to reach to the goal the turtle waffle pi is used to make the trajectory in Gazebo to reach the desired point while avoiding the obstacle either the obstacle it do find in Gazebo is static or dynamic that all depends on the situation but have to be visualize to these robot have make their complete trajectory reach to their goal.

Chapter 3

Pedestrian dynamic modeling

3.1 Introduction

Pedestrian Dynamic Modeling completely covers the behaviour of pedestrians in the the road like scenario after this pedestrian behaviours need to understand by the robots to like a human being, for this different Machine learning methods is to be applied to under the pedestrian dynamics, covering all this scenario, it is to be seen that we need to undrestand to give the viualizualize through the help of either understanding the human cohesion or understanding their complete social factors like person is individual or more than a a single person is there and trying to make the pedestrian to identified with some name so that these individuals can be track throughout, and understand their velocity so that understanding the pattern trajectory can be easily understand further knowing these all factors also addition of group size also understanding their interaction cohesion score to understand the pedstrian dynamic modelling, Below section try to cover all these areas one by one.

3.2 Social cohesion analysis

Social cohesion Analysis constitute fundamental components of socially intelligent robotic navigation systems operating in dynamic and crowded environments. Traditional robot navigation approaches often consider pedestrians as moving obstacles and focus primarily on collision avoidance. However, in real-world environments, humans exhibit complex social behaviors, move in groups, maintain interpersonal relationships, and follow social norms. Therefore, a socially intelligent robot must not only detect the presence of pedestrians but also understand their interactions, movement patterns, and group structures to navigate in a manner that is both safe and socially acceptable. The first step in this process is human detection, which enables the robot to perceive pedestrians in its surroundings and understand its behavior. In this work, YOLO (You Only Look Once) is employed for real-time human detection due to its high detection accuracy and computational efficiency to track people or other obstacle easily. YOLOv5 processes each camera frame by dividing the image into grids and predicting bounding boxes along with class probabilities in a single forward pass. This architecture allows the system to detect multiple pedestrians simultaneously while maintaining real-time performance, making it suitable for robotic applications operating in dynamic environments. The detected pedestrians are human detection and cohesion modeling are starting for components of socially intelligent robotic navigation systems operating in dynamic and crowded environments. For robots to navigate safely and efficiently alongside pedestrians, they must not only detect the presence of individuals but also understand the social relationships and group dynamics that influence human movement while making trajectory and also help in approaching the human beings, 3d estimation. In socially intelligent navigation, understanding human cohesion allows robots to generate paths that respect personal space and preserve group integrity. Instead of treating pedestrians as independent obstacles, the robot considers social structures and adapts its behavior accordingly. This approach improves navigation efficiency, enhances human comfort, and promotes safer human-robot interaction in environments such as airports, shopping malls, hospitals. The integration of human detection, tracking, group identification, and cohesion modeling forms the foundation of advanced social navigation frameworks, enabling robots to operate naturally and acceptably within human-centered environments. To ensure socially intelligent navigation, the robot must be aware of human presence in its environment. This is achieved through real-time human detection using YOLOv5, a state-of-the-art object detection algorithm known for its speed and accuracy. YOLOv5: YOLOv5 is a single-stage object detector that processes the entire image at once and directly predicts bounding boxes and class probabilities. Human

Detection Workflow

1. 1.Input: The robot's onboard RGB camera continuously captures video frames of the surrounding environment.

2. 2.Detection: YOLO is applied to each frame to detect objects. Specifically, it identifies humans by classifying them with a dedicated 'person' label. Each detected human is enclosed in a bounding box, which provides the pixel coordinates of the human's position in the image.

3. 3.Centroid Extraction: These centroids help in understanding the spatial distribution of people in the environment.

Why This Matters: The bounding boxes and centroids serve as spatial references for the navigation system. This spatial information is fed into further modules (like DeepSORT for tracking and potential field algorithms for path planning) to allow the robot to avoid invading human personal space and to make socially compliant navigation decisions.

3.2.1 Cohesion modeling scores analysis

Strategy proximity-based grouping: Individuals within a threshold distance are grouped. Group cohesion is assessed using the following metrics:

Proximity cohesion score

Grouping individuals by calculating pairwise distances and clustering those within a predefined threshold. Compute spatial distance between detected individuals using their positions. Assign individuals to the same group if their distance is below the threshold. Measure how close group members are to each other,based on the average distance between individuals in the group. [21] equation(3.1)

$$C_p(t) = K_p \cdot \sum_{i=1}^{N} \sum_{j=i+1}^{N} \frac{1}{\text{dist}(i,j)} \tag{3.1}$$

Where:

- $C_p(t)$: Proximity cohesion score for the group at time t

- K_p : Normalization constant
- N : Number of individuals in the group
- $\text{dist}(i, j)$: Euclidean distance between member i and member j

The Proximity cohesion score is calculated using Equation (3.1). Similar proximity-based social interaction measures have been widely used in pedestrian group behavior and crowd dynamics studies [8, 18, 21].

Walking speed cohesion score

Measure how similar the walking speeds of individuals are within a group ,reflecting synchronized movement.Equation (3.2) [21]

$$1001[21]C_w(G^{(k,t)}) = K_w \cdot \frac{\left(\frac{\sum_{i \in ID^t} ||V^i||}{N}\right)}{\left(\frac{\sum_{j \in G^{(k,t)}} ||V^j||}{n^k}\right)} \tag{3.2}$$

Where:

- $C_w(G^{(k,t)})$: Walking Speed Cohesion Score for group G_k at time t
- K_w : Normalization constant
- $||V^i||$: Walking speed of individual i
- ID^t : Set of all detected individuals at time t
- N : Total number of detected individuals
- $G^{(k,t)}$: Group k at time t
- $||V^j||$: Walking speed of group member j
- n^k : Number of members in group G_k

The walking speed cohesion score is computed using Equation (3.2).The metric evaluates the similarity of pedestrian walking speeds and reflects the degree of synchronized group movement [8, 14, 16].

Group size cohesion score

Reflects the inherent cohesion of a group of a group based on its size, assuming larger groups exhibit stronger collective behavior and stability. Cohesion increases with group size(nkn , knk), making larger groups appear less penetrable and more cohesive. Group size cohesion reflects the inherent cohesion of a pedestrian group based on its size. Larger groups are generally considered to exhibit stronger collective behavior, greater stability, and lower penetrability compared to smaller groups. Therefore, the cohesion score increases with the number of members in the group.Equation (3.3) [21]

$$1001[21]C_s(G^{k,t}) = K_s \cdot n^k \tag{3.3}$$

where:

- $C_s(G^{k,t})$: Group size cohesion score of group G_k at time t.
- n^k : Number of members in group G_k.
- K_s : Normalization constant used to scale the cohesion score.

The Group Size Cohesion Score is determined using Equation (3.3). Larger pedestrian groups generally exhibit stronger social cohesion, collective behavior, and greater group stability than smaller groups [8, 18, 21]. Larger pedestrian groups generally exhibit stronger collective behavior, enhanced stability, and lower penetrability than smaller groups. The formulation assumes that cohesion increases proportionally with group size, implying that larger pedestrian groups tend to maintain stronger social bonds and coordinated movement patterns than smaller groups.

Interaction cohesion score

Measures group cohesion based on level of mutual engagement and attention among individuals, reflected through their 3Dface orientation vectors.Equation(3.4) [21]

$$1001[21]C_i(G^{(k,t)}) = K_i \cdot \frac{1}{n^k} \sum_{\substack{i \neq j \\ i,j \in G^{(k,t)}}} \frac{\sin(\theta_{i,j})}{\cos(\theta_{i,j})} \quad \text{with } \theta_{i,j} \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \tag{3.4}$$

Where:

- $C_i(G^{(k,t)})$: Interaction cohesion score for group G_k at time t
- K_i : Normalization constant
- n^k : Number of members in group G_k
- i, j : Individual members in group $G^{(k,t)}$, with $i \neq j$
- $\theta_{i,j}$: Angle between 3D face orientation vectors f_i^0 and f_j^0 in the X-Y plane of the camera coordinate system
- $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$: Valid angular range for $\theta_{i,j}$

The interaction cohesion score is calculated using Equation (3.4). The score measures mutual engagement among group members through face orientation and interaction cues, which are important indicators of social relationships within pedestrian groups [9, 18, 21].

Overall cohesion score

Equation (3.5) [21]

$$C(t) = C_p(t) + C_i(t) + C_s(t) \tag{3.5}$$

Where:

- $C(t)$: Total group cohesion score at time t
- $C_p(t)$: Proximity cohesion score
- $C_i(t)$: Interaction cohesion score
- $C_s(t)$: Group size cohesion score

The Overall Cohesion Score is obtained using Equation (3.5), which combines proximity, interaction, and group-size cohesion components to provide a comprehensive measure of group cohesiveness. This integrated measure provides a comprehensive representation of pedestrian group cohesiveness [18, 21].

3.2.2 Implementation of social modeling score

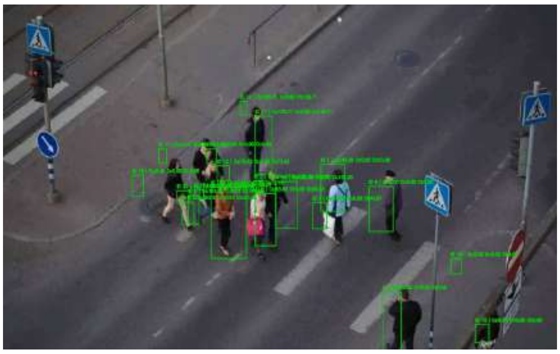

Figure 3.1: Simulation snap of visualization of pedestrian with cohesion

Fig3.1 Figure here shown for all the pedestrian surrounding and have to shown their visualization by showing their respective cohesion score and visualize with showing the values in the respective between frames to have visualize corresponding cohesion score, proximity cohesion score, group size score, walking speed score and overall cohesion score.

3.2.3 Implementation of cohesion scores in Gazebo

This is the implementation practice in Gazebo, how it do behave that simulation have to be used to perform in YOLO and predicting the pedestrian position in this Gazebo,we are trying to predict in Rviz to visualize this pedestrian. The simulation environment is first created in Gazebo, where multiple pedestrian models move dynamically along predefined or random paths. These pedestrians represent humans walking in real-world environments such as corridors, malls, or public spaces. A camera sensor mounted on the robot continuously captures images of the surrounding environment and publishes them through ROS topics. The captured images are processed using the YOLO (You Only Look Once) object detection algorithm. YOLO detects pedestrians in each frame and provides bounding box coordinates along with confidence scores. Since detection alone does not provide continuous motion information, DeepSORT is integrated with YOLO to track pedestrians across consecutive frames. DeepSORT assigns unique IDs to each pedestrian and generates their movement trajectories over time.

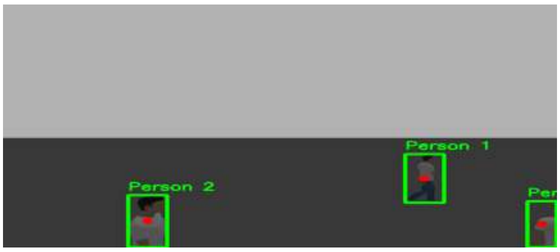

Figure 3.2: Simulation snap of detect of 3 pedestrianas in Gazebo

Fig3.2: Here three pedestrians is detected using YOLO performed inside the Gazebo . This implementation of finding pedestrian in this gazebo space to make the detection for visualization of detected pedestrians and take help of this data to see in Rviz.

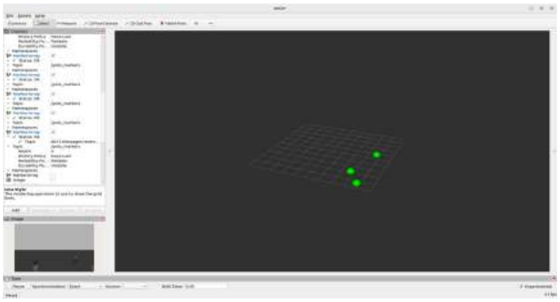

Figure 3.3: Simulation snap of visualization of detection of pedestrian in Rviz extracted from Gazebo

Fig3.3: Here three pedestrians is detected using YOLO performed inside the Gazebo is extracted to interface in RVIZ that three dot is giving the indication these three pedestrians .

Figure 3.4: Simulation snap of multiple person surrounding and 1 waffle pi robot in the middle in Gazebo

Fig3.4Here Multiple pedestrians is inside the Gazebo , and a waffle pi Robot is there to demonstrate the presence of pedestrians through the waffle pi using yolo to detect the person and perform movement.Here few pedestrians surrounding the space in Gazebo , giving the interface like real time some of the static pedestrian there and some of the dynamic pedestrian visualization.

Figure 3.5: Simulation snap of detection of multiple pedestrians using YOLO

Fig 3.5: Here Multiple pedestrians is inside the Gazebo , and a waffle pi Robot is there to demonstrate the presence of pedestrians through the waffle pi using yolo to detect the person, so detection is determined and show in bounding purple box of all the pedestrian by the camera attach in waffle pi and perform movement.

3.3 Pedestrian modeling analysis

Socially friendly navigation framework for good stability and expansibility in the background of human-aware navigation of service robot, which is mainly divide in two parts :

1. Pedestrian modeling and trajectory prediction: This module models the individual space and group interaction of pedestrians based on the posture and motion information of pedestrians detected globally. Then, a multilayer social cost map containing future pedestrian posture and group interaction information generated through the pedestrian trajectory prediction network.

2. Global path planner: This module combines the global static cost map, the local obstacle cost map, and the multilayer social cost map to provide constraints for path planning, and uses the "plan-prediction-execution" cycle for dynamic path planning. Before the robot reaches the target pose, the time is divided into several equal planning cycles.

3.3.1 Overview of different social factors for pedestrian modeling

Personal comfort distance modelling

Pedestrian i as a cost function p costi(x,y) of asymmetric Gaussian distribution based on the proximity model and pedestrian speed information as the comfort space of pedestrians is directly proportional to the speed of pedestrians.Equation(3.6),(3.7) [21]

$$1001[21]p^{i}_{\text{cost}}(x,y) = A \cdot \exp\left(-\left(\frac{d \cdot \cos\theta}{2\sigma_x}\right)^2 - \left(\frac{d \cdot \sin\phi}{2\sigma_y}\right)^2\right) \tag{3.6}$$

$$1001[21]d = \sqrt{(x-x_i)^2 + (y-y_i)^2}, \quad \phi = \arctan 2(y-y_i, x-x_i) - \phi_i \tag{3.7}$$

The pedestrian personal-space cost function represented by Equations (3.6) and (3.7) is based on the anisotropic Gaussian model used in socially aware navigation to represent human comfort zones and personal space around pedestrians based on the anisotropic Gaussian model used in socially aware navigation to represent human comfort zones and personal space around pedestrians. The model accounts for pedestrian position, orientation, and walking direction to generate an asymmetric comfort field around each individual [4, 5, 22].

Group interaction detection

Detect pedestrian groups based on the pedestrian position and direction information, and represent each detected pedestrian with a continuous function of Gaussian distribution.Equation(3.8) [21]

$$f_i(x,y) = \exp\left(-a(x-\bar{x}_i)^2 - 2b(x-\bar{x}_i)(y-\bar{y}_i) + c(y-\bar{y}_i)^2\right) \tag{3.8}$$

Equation (3.8) models each pedestrian using a Gaussian distribution to facilitate group interaction detection and social relationship analysis among pedestrians. Gaussian-based representations have been widely used for identifying pedestrian interactions and estimating social grouping behavior in crowded environments [18, 21].

Group interaction modelling

We use the least square method to fit the minimal circle of the detected pedestrian group members, and set the social cost gcosti(x,y) to reflect the cost of the mobile robot passing through the interacting pedestrian group. Equation(3.9),(3.10) [21]

$$g_{\cos}^{t_i}(x,y) = \begin{cases} 1001[21]\exp\left(-\left(\left(\frac{d}{2\sigma_x}\right)^2 + \left(\frac{d}{2\sigma_y}\right)^2\right)\right), & \text{if } d > R \end{cases} \tag{3.9}$$

$$d = (x-x_i)^2 + (y-y_i)^2 \tag{3.10}$$

The group interaction cost model described by Equations (3.9)and (3.10) assigns a social navigation cost to pedestrian groups and discourages robots from penetrating socially interacting groups [18, 21].. Algorithm of pedestrian modeling: To understand motion patterns, group behaviors, and intention of pedestrians over time to predict future trajectories and adapt robotic navigation accordingly. Human Detection + Kalman Filter + YOLOv5 +Deep SORT + Python. These help to traced human being and counts number of human beings in a Frame with uniqueID and their respective velocity and estimate their future trajectories. Algorithm of work done

- STEP 1. Pedestrian detection and tracking: This step identifies and follows pedestrians across frames in a video: Each frame is processed using YOLOv5. The detected bounding boxes are passed to Deep SORT, which assigns a unique ID to each pedestrian and maintains that identity over time.

- STEP 2. Velocity Estimation: This step estimates how fast each pedestrian is moving . The center point of each pedestrian's bounding box is calculated for each pair of consecutive frames, the Euclidean distance moved is computed and divided by the time difference to estimate speed.This gives a time series of velocity values for every pedestrian.

Steps for speed calculation:

1. **Center Point of bounding box**

   If a pedestrian's bounding box is given by: [10, 11]. Top-left corner: $(x_{\min}, y_{\min})$, Bottom-right corner: $(x_{\max}, y_{\max})$ The center point of the bounding box is: Equation(3.11) [10].

   $$(X_c, Y_c) = \left(\frac{x_{\min} + x_{\max}}{2}, \frac{y_{\min} + y_{\max}}{2} \right) \tag{3.11}$$

   The center point calculation is performed using equation(3.11) to represent the pedestrian location within the image frame based on object detection outputs generated by YOLO and tracking frameworks [10, 11].

2. **Euclidean distance between consecutive frames**

   For the same pedestrian at times t_1 and t_2, let (X_1, Y_1) and (X_2, Y_2) be the center points. The distance moved is: Equation(3.12) [11].

   $$d = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2} \tag{3.12}$$

   Equation (3.12) computes the displacement of a pedestrian between consecutive frames using the Euclidean distance metric which forms the basis for velocity estimation in multi-object tracking systems [10, 11].

3. **Velocity (Speed) Estimation [10, 11].**

   Given time difference $\Delta t = t_2 - t_1$, the velocity is: Equation(3.13) [11]

   $$V = \frac{d}{\Delta t} \tag{3.13}$$

The pedestrian velocity is estimated using Equation (3.11), which calculates the displacement per unit time from tracked pedestrian positions which calculates the displacement per unit time from tracked pedestrian positions [10, 11]. This initially gives speed in *pixels per second*.

- STEP 3.Future trajectory prediction : Estimation of each pedestrian's past positions is used to extrapolate future positions using linear interpolation. The predicted future positions are compared to actual future positions to calculate trajectory prediction error (distance between predicted and actual). Smoothing (like a moving average) is applied to remove noise in error curves.

- STEP 4. Group detection(Clustering pedestrians) : Find groups of pedestrians who are walking close together. At each frame, the (x, y) positions of pedestrians are clustered using a method DBSCAN (Density based SCAN(Spatial clustering of application with noise). Pedestrians assigned to the same cluster are considered part of the same social group. Each pedestrian is labeled with a group ID.

- STEP 5. Cohesion score calculation : This stage quantifies the social cohesion within each pedestrian group. Proximity cohesion : Measures how physically close group members are to each other. Smaller average distances mean higher cohesion. Velocity C=cohesion : Measures how similarly group members are moving. A low standard deviation in speed and direction indicates cohesive movement. Interaction cohesion (if face orientation is available) : Measures on the basis of facing orientation pedestrians toward each other in a group . Calculated by comparing face orientation angles between members. All three scores can be normalized to [0,1] and combined to form a total cohesion score.

- STEP 6. Visualization : This step overlays information on the video for analysis and interpretation. Bounding boxes are shown with pedestrian IDs. Predicted future trajectories (dashed lines),On-screen cohesion score

- STEP 7. Training, Loss curves, and evaluation : Training a model for future prediction, this step monitors performance: Loss functions (MSE between predicted and actual trajectory) are computed. Cohesion-aware loss functions can be added to predictions. Smoothed loss curves are plotted to observe learning progress. Early stopping criteria (e.g., no improvement after N=9 epochs) can halt training when optimal and criteria of ML model stopping.

3.3.2 Performance analysis of cohesion algorithm

Pedestrian tracking

- Algorithm: YOLOv5 + Deep Sort,
- Performance (Qualitative): Tracking appears stable for most pedestrians in the crossing scene.
- IDs are mostly consistent across frames, enabling grouping and velocity computation.
- Limitations: Occasional ID switches or missed detections when pedestrians are occluded or close.
- Improvement Areas: Use better re-ID features or optical flow integration.

Velocity estimation

- Method: Frame-wise center position change per pedestrian.
- Results: Real-time estimated velocity curves show smooth motion for most individuals.
- Captures both slow and fast walkers.
- Observations:Sudden spikes in velocity likely due to tracking noise or frame skip.
- Suggestion:Apply Kalman smoothing or optical flow refinement.

Future trajectory prediction

:Linear interpolation baseline (frame-wise)

- Evaluation (Raw prediction error): Spikes in frames 150–220, likely due to trajectory changes or tracking drift.

- Smoothed Error (with moving average): Reduces noise and shows overall good prediction in stable sections.
- Implication:Linear interpolation works in short-term, but errors grow over longer horizons or group interactions.
- Next step: Train a learning model (e.g., LSTM or Transformer) using extracted trajectory data for better accuracy.

Cohesion scoring

- Metrics: Proximity cohesion – how close pedestrians walk together.
- Velocity Cohesion – similarity in walking speed.,Interaction Cohesion – directional alignment (from face orientation
- Visual Results:Cohesion heatmaps & plots show,Some groups (2–3 pedestrians) have high cohesion scores.,Sparse or ungrouped pedestrians have low cohesion.
- Conclusion: Individuals with social interaction are clearly identified.

3.3.3 Implementation of pedestrian modeling

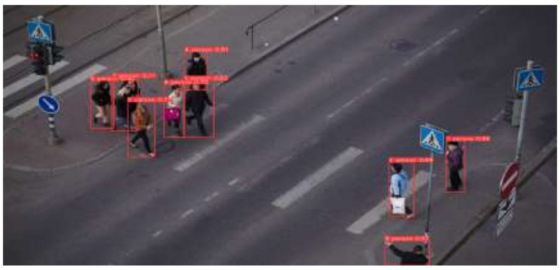

Figure 3.6: Simulation snap of performed Deep SORT on multiple people

Fig 3.6In this figure Counting people provided unique ID with estimation of their Respective size frame Pedestrian modeling and trajectory prediction. This module models the individual space and group interaction of pedestrians based on the posture and motion information of pedestrians detected globally.Then, a multilayer social cost map containing future pedestrian posture and group interaction information is generated through the pedestrian trajectory prediction network

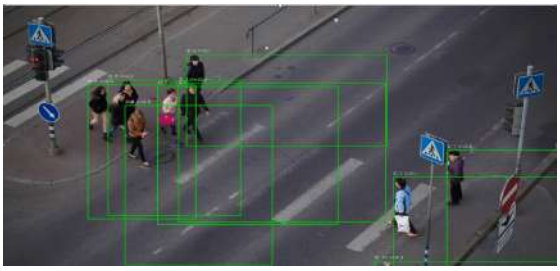

Figure 3.7: Simulation snap of Pedestrian dynamic with velocity and future trajectory

Fig 3.7Here all pedestrian are having their walking speed so specifies each walking speed with frame ID .To Understand motion patterns, group behaviors, and intention of pedestrians over time to predict future trajectories and adapt robotic navigation accordingly. Human Detection + Kalman Filter + YOLOv5 +Deep SORT + Python. These help to traced human being and counts number of human beings in a Frame with unique ID and their respective velocity and estimate their future trajectories

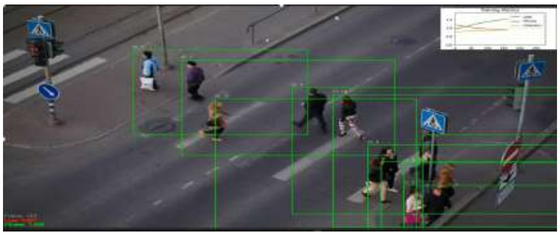

Figure 3.8: Simulation snaps of training of Plot for fitness, frame loss

Fig3.8 :In This figure, these pedestrians are crossing road after tracing their ID and velocity calculation from past trajectory it understand slowly their future trajectory so here in this frame At every frame there fitnesses are showing and losses are showing in left corner and right top their plotting is present. Per Frame Loss, fitness and its plotting performed training metricies.

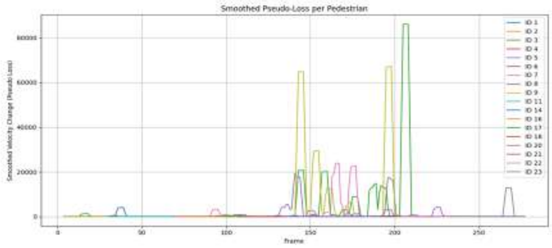

Figure 3.9: Simulation snap of plot of smoothed pseudo loss per pedestrian

Fig3.9Here the plotting is in between smoothed velocity change (pseudo loss of every pedestrian in per frame. Basically at every frame that is changing every second so what robot was expecting velocity and what actual velocity difference is showing)

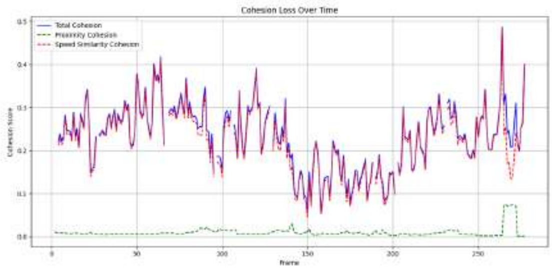

Figure 3.10: Simulation snap of plot of cohesion loss over time

Fig 3.10 Here cohesion score per frame is showing blur is showing Total cohesion, dotted green is showing proximity cohesion , dotted red is showing speed similarity cohesion

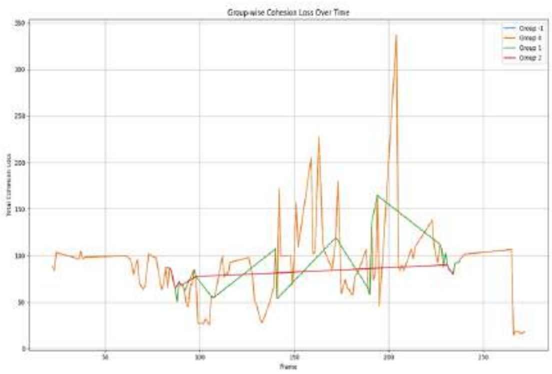

Figure 3.11: Simulation snap of groupwise cohesion score over time

Fig 3.11Here four groups are showing in 4 different colours ang total cohesion loss over time time plotting is shown. This shows how do loss of cohesion changes with time stamp overall grouping understand of the cohesion.

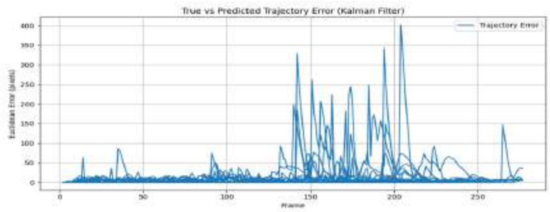

Figure 3.12: Simulation snap of true vs predicted trajectory error

Fig 3.12Here true vs predicted error on per frame shown in the middle its become very high because a unpredicted pedestrian enter so loss increases slowly slowly settle. This variation helps to understand the trajectory and help to understand the pattern.

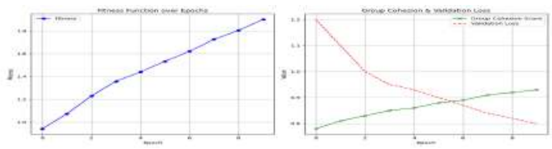

Figure 3.13: Simulation snap of fitness function over epochs, group cohesion and validation loss

Fig 3.13 In figure as time passes become better so graph increases and figure 2 group cohesion green color increase and loss red color decrease .How it fit as prediction so this help to find the accurate trajectory in the future.

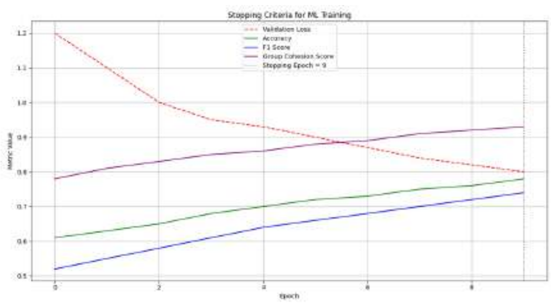

Figure 3.14: Simulation of plot of stopping criteria for ML

Fig3.14 Here ML is applied after training that when losses get increase its better to stop than again making forward and f1 score and accuracy shows.This accuracy shows how accurate trajectory is being made.

3.4 Pedestrian trajectory analysis

Pedestrian trajectory is a fundamental component of socially aware robot navigation, enabling robots to understand, predict, and adapt to human movement patterns in dynamic environments. A pedestrian trajectory represents the sequence of positions occupied by a person over time and provides valuable information about their direction, velocity, acceleration, and future intentions. By analyzing these trajectories, robots can anticipate pedestrian motion and make navigation decisions that are safe, efficient, and socially acceptable. Their motion is influenced by surrounding people, obstacles, group interactions, and social norms. Therefore, robots must not only detect pedestrians but also predict their future trajectories to avoid collisions and minimize social discomfort. Trajectory modeling helps robots understand crowd dynamics, identify walking patterns, and estimate future pedestrian positions, thereby enabling proactive navigation.

3.4.1 Implementation of Gaussian Membership functions for direction estimation

Path Length

Equations (3.14),(3.15), (3.16) [13]

$$\mu_{SL}(L) = e^{-\frac{L^2}{1500}} \tag{3.14}$$

$$citeref713\mu_{ML}(L) = e^{-\frac{(L-750)^2}{1500}} \tag{3.15}$$

$$citeref713\mu_{LL}(L) = e^{-\frac{(L-1500)^2}{1500}} \tag{3.16}$$

Pedestrian density

Equations (3.17), (3.18) , (3.19) [13]

$$citeref13\mu_{S\rho}(\rho) = e^{-0.7\rho^2} \tag{3.17}$$

$$\mu_{M\rho}(\rho) = e^{-0.7(\rho-2)^2} \tag{3.18}$$

$$citeref13\mu_{L\rho}(\rho) = e^{-0.7(\rho-4)^2} \tag{3.19}$$

Road width

Equations (3.20), (3.21), (3.22) [13]

$$\mu_{SW}(W) = e^{-\frac{W^2}{7.5}} \tag{3.20}$$

$$\mu_{MW}(W) = e^{-\frac{(W-7.5)^2}{7.5}} \tag{3.21}$$

$$\mu_{LW}(W) = e^{-\frac{(W-15)^2}{7.5}} \tag{3.22}$$

Open space

Equations (3.23), (3.24), (3.25) [13]

$$\mu_{SR}(R) = e^{-11R^2} \tag{3.23}$$

$$citeref713\mu_{MR}(R) = e^{-11(R-0.5)^2} \tag{3.24}$$

$$citeref713\mu_{LR}(R) = e^{-11(R-1)^2} \tag{3.25}$$

Fuzzy Rule Structure

$$\text{IF } L \text{ is } SL \text{ AND } \rho \text{ is } S\rho \text{ AND } W \text{ is } LW \text{ AND } R \text{ is } LR \text{ THEN } P \text{ is } BP \tag{3.26}$$

Path Selection Mapping

Equations (3.27), (3.28) [13]

$$(L_k, \rho_k, W_k, R_k) \Rightarrow P_k \tag{3.27}$$

$$P_k = \text{Probability that pedestrian selects path } k \tag{3.28}$$

Output Fuzzy Set

Equation (3.29) [13]

$$P \in \{SP,\ LSP,\ MP,\ LBP,\ BP\} \tag{3.29}$$

Consequent-oriented fuzzy interference

Equations (3.31), (3.30), (3.32) [13]

$$U^{out}(P) = \sum_i \frac{\alpha_L \mu_{iL}(L_0) + \alpha_\rho \mu_{i\rho}(\rho_0) + \alpha_W \mu_{iW}(W_0) + \alpha_R \mu_{iR}(R_0)}{\sum_k \alpha_k} \cdot \mu_{iP}(P) \tag{3.30}$$

$$\alpha_L, \alpha_\rho, \alpha_W, \alpha_R \text{ are the rule weights} \tag{3.31}$$

$$\alpha_L > \alpha_\rho, \alpha_W, \alpha_R \tag{3.32}$$

(Path length has the highest width)

Defuzzification (Final numeric probability)

Equations (3.33), (3.34), (3.35), (3.36), (3.37), (3.36), (3.38) [13]

$$P_k = \frac{\int_0^1 P\,\mu(P)\,dP}{\int_0^1 \mu(P)\,dP} \tag{3.33}$$

$$P_{left},\ P_{center},\ P_{right} \in [0,1] \tag{3.34}$$

Final decision rule

$$CHOSENPATH = \arg \max_{k\in\{L,C,R\}} P_k \tag{3.35}$$

Movement decision condition

$$\text{LEFT if } P_{left} > P_{center} \text{ and } P_{left} > P_{right} \tag{3.36}$$

$$\text{CENTER if } P_{center} = \max(P_{left}, P_{center}, P_{right}) \tag{3.37}$$

$$\text{RIGHT if } P_{right} = \max(P_{left}, P_{center}, P_{right}) \tag{3.38}$$

3.4.2 Implementation of direction estimation with an example using Gaussian Function

We consider one pedestrian and one decision instant. The crisp inputs to the fuzzy system are chosen as

$$L_0 = 300, \qquad \rho_0 = 1.0, \qquad W_0 = 5, \qquad R_0 = 0.3,$$

where L is the path length, ρ is the local density,W is the road width and R is the open space.

1 Fuzzification using Gaussian membership functions The Gaussian membership functions used in this work are , all formula extracted from equations Equations (3.39), (3.40), (3.41), (3.42), (3.43), (3.44), (3.45), and (3.46) [13]:

Fuzzification Using Gaussian Membership Functions

$$\mu_{SL}(L) = e^{-\frac{L^2}{1500}}, \qquad \mu_{ML}(L) = e^{-\frac{(L-750)^2}{1500}}, \qquad \mu_{LL}(L) = e^{-\frac{(L-1500)^2}{1500}} \tag{3.39}$$

$$\mu_{Sp}(\rho) = e^{-0.7\rho^2}, \qquad \mu_{Mp}(\rho) = e^{-0.7(\rho-2)^2}, \qquad \mu_{Lp}(\rho) = e^{-0.7(\rho-4)^2} \tag{3.40}$$

$$\mu_{SW}(W) = e^{-\frac{W^2}{7.5}}, \qquad \mu_{MW}(W) = e^{-\frac{(W-7.5)^2}{7.5}}, \qquad \mu_{LW}(W) = e^{-\frac{(W-15)^2}{7.5}} \tag{3.41}$$

$$\mu_{SR}(R) = e^{-11R^2}, \qquad \mu_{MR}(R) = e^{-11(R-0.5)^2}, \qquad \mu_{LR}(R) = e^{-11(R-1)^2} \tag{3.42}$$

Substituting $L_0 = 300$, $\rho_0 = 1.0$, $W_0 = 5$, and $R_0 = 0.3$, the membership values are

$$\mu_{SL}(L_0) \approx 0.14, \qquad \mu_{ML}(L_0) \approx 0, \qquad \mu_{LL}(L_0) \approx 0 \tag{3.43}$$

$$\mu_{Sp}(\rho_0) \approx 0.50, \qquad \mu_{Mp}(\rho_0) \approx 0.50, \qquad \mu_{Lp}(\rho_0) \approx 0.02 \tag{3.44}$$

$$\mu_{SW}(W_0) \approx 0.001, \qquad \mu_{MW}(W_0) \approx 0.46, \qquad \mu_{LW}(W_0) \approx 0.04 \tag{3.45}$$

$$\mu_{SR}(R_0) \approx 0.37, \qquad \mu_{MR}(R_0) \approx 0.91, \qquad \mu_{LR}(R_0) \approx 0.30 \tag{3.46}$$

Hence, the dominant fuzzy labels are $L \approx SL$, $\rho \approx Sp/Mp$, $W \approx MW$, $R \approx MR$.

2) Rule activation using the rule matrix From the fuzzy rule base (Table I), consider the following three rules: [13]

Rule L: IF L is SL, ρ is Sp, W is MW, R is $MR \Rightarrow P = BP$ (LEFT),

Rule C: IF L is SL, ρ is Mp, W is MW, R is $MR \Rightarrow P = MP$ (CENTER),

Rule R: IF L is SL, ρ is Lp, W is MW, R is $LR \Rightarrow P = SP$ (RIGHT).

Using the minimum operator for conjunction, the firing strengths are [13]

$$\mu_{\mathrm{L}} = \min\big(\mu_{SL}(L_0), \mu_{Sp}(\rho_0), \mu_{MW}(W_0), \mu_{MR}(R_0)\big) = \min(0.14, 0.50, 0.46, 0.91) = 0.14,$$

$$\mu_{\mathrm{C}} = \min\big(\mu_{SL}(L_0), \mu_{Mp}(\rho_0), \mu_{MW}(W_0), \mu_{MR}(R_0)\big) = 0.14,$$

$$\mu_{\mathrm{R}} = \min\big(\mu_{SL}(L_0), \mu_{Lp}(\rho_0), \mu_{MW}(W_0), \mu_{LR}(R_0)\big) = \min(0.14, 0.02, 0.46, 0.30) = 0.02.$$

Thus, Rule L and Rule C are strongly active, whereas Rule R is weak.

3.Consequent-oriented fuzzy inference Using the consequent-oriented fuzzy inference scheme [13], the aggregated output corresponding to the i^{th} rule is computed as The consequent-oriented fuzzy inference mechanism is described by Equations (3.47)–(3.51) [13].

$$U_i^{out}(P) = \frac{\alpha_L \mu_{iL}(L_0) + \alpha_\rho \mu_{i\rho}(\rho_0) + \alpha_W \mu_{iW}(W_0) + \alpha_R \mu_{iR}(R_0)}{\sum_k \alpha_k} \mu_{iP}(P) \tag{3.47}$$

where α_L, α_ρ, α_W, and α_R denote the input weights. In this work,

$$\alpha_L = 0.4, \qquad \alpha_\rho = \alpha_W = \alpha_R = 0.2 \tag{3.48}$$

indicating that path length has the highest influence on the final decision.

For the LEFT path, the corresponding rule weight is obtained as

$$\omega_{\mathrm{L}} = \frac{0.4\,\mu_{SL}(L_0) + 0.2\,\mu_{Sp}(\rho_0) + 0.2\,\mu_{MW}(W_0) + 0.2\,\mu_{MR}(R_0)}{1.0} \tag{3.49}$$

and the output membership function becomes

$$U_{\mathrm{L}}^{out}(P) = \omega_{\mathrm{L}} \mu_{BP}(P) \tag{3.50}$$

Similarly,

$$U_{\mathrm{C}}^{out}(P) = \omega_{\mathrm{C}} \mu_{MP}(P), \qquad U_{\mathrm{R}}^{out}(P) = \omega_{\mathrm{R}} \mu_{SP}(P) \tag{3.51}$$

The overall fuzzy output for each candidate path $k \in \{L, C, R\}$ is obtained by aggregating the contributions of all activated rules corresponding to that path.

4. Defuzzification (centroid method) For each candidate path $k \in \{L, C, R\}$, the final numeric probability P_k is obtained by centroid defuzzification: The centroid defuzzification process is given by Equation (3.52). The output fuzzy set centers are defined in Equation (3.53), while the resulting path-selection probabilities are obtained from Equation (3.54). [13]

$$P_k = \frac{\int_0^1 P\,\mu_k(P)\,dP}{\int_0^1 \mu_k(P)\,dP} \approx \frac{h^k_{\mathrm{SP}} c_{\mathrm{SP}} + h^k_{\mathrm{MP}} c_{\mathrm{MP}} + h^k_{\mathrm{BP}} c_{\mathrm{BP}}}{h^k_{\mathrm{SP}} + h^k_{\mathrm{MP}} + h^k_{\mathrm{BP}}} \tag{3.52}$$

$$c_{\mathrm{SP}} = 0.1, \qquad c_{\mathrm{MP}} = 0.5, \qquad c_{\mathrm{BP}} = 0.9 \tag{3.53}$$

$$P_{\mathrm{left}} \approx 0.73, \qquad P_{\mathrm{center}} \approx 0.50, \qquad P_{\mathrm{right}} \approx 0.10, \qquad P_k \in [0,1] \tag{3.54}$$

where $\mu_k(P)$ is the aggregated membership function of the output fuzzy sets (SP, LSP, MP, LBP, BP) for path k.For implementation it is convenient to approximate the centroid using the centres [13] All three values satisfy $P_k \in [0,1]$.

5. Final decision rule The actual pedestrian decision is taken by selecting the path with the maximum probability: equation (3.55) [13]

$$\text{CHOSENPATH} = \arg \max_{k \in \{L,C,R\}} P_k = \arg\max(0.73,\ 0.50,\ 0.10) = \text{LEFT} \tag{3.55}$$

Thus, for the input state (L_0, ρ_0, W_0, R_0), the fuzzy system predicts that the pedestrian will move toward the left path [13]. The Gaussian membership functions defined in equations (3.14) to (3.15)are used to represent the uncertainty associated with path length, pedestrian density, road width, and open space. Similar fuzzy inference based path-selection models have been employed in pedestrian movement prediction and crowd behavior analysis [9, 13].

The fuzzy rule structure described in Equation (3.20) establishes the relationship between environmental parameters and pedestrian path preferences. Rule-based decision systems have been widely used for modeling pedestrian navigation behavior under uncertain conditions [9,16]. The path selection mapping represented by Equation (3.33) converts environmental observations into probabilistic path choices and facilitates trajectory estimation for pedestrian movement prediction [9, 19].

The consequent-oriented fuzzy inference mechanism shown in Equation (3.30) aggregates the contribution of multiple fuzzy rules and computes the overall output membership function. Similar weighted fuzzy inference approaches have been utilized for human behavior modeling and navigation decision making [9, 13].

The centroid-based defuzzification process represented by Equation (**??**) converts the aggregated fuzzy output into a crisp probability value for each candidate path. Centroid defuzzification is one of the most commonly adopted methods in fuzzy decision systems due to its stability and smooth output characteristics [9].

Table 3.1: Fuzzy Rule Base

L	ρ	W	R	P
SL	Sp	SW	SR	BP
SL	Sp	SW	MR	BP
SL	Sp	SW	LR	BP
SL	Sp	MW	SR	BP
SL	Sp	MW	MR	BP
SL	Sp	MW	LR	BP
SL	Sp	LW	SR	BP
SL	Sp	LW	MR	BP
SL	Sp	LW	LR	BP
SL	Mp	SW	SR	LBP
SL	Mp	SW	MR	LBP
SL	Mp	SW	LR	BP
SL	Mp	MW	SR	BP
SL	Mp	MW	MR	BP
SL	Mp	MW	LR	BP
SL	Mp	LW	SR	BP
SL	Mp	LW	MR	BP
SL	Mp	LW	LR	BP
SL	Lp	SW	SR	MP
SL	Lp	SW	MR	MP
SL	Lp	SW	LR	LBP
SL	Lp	MW	SR	LBP
SL	Lp	MW	MR	LBP
SL	Lp	MW	LR	LBP

L	ρ	W	R	P
ML	Sp	SW	SR	MP
ML	Sp	SW	MR	MP
ML	Sp	SW	LR	MP
ML	Sp	MW	SR	MP
ML	Sp	MW	MR	MP
ML	Sp	MW	LR	LBP
ML	Sp	LW	SR	MP
ML	Sp	LW	MR	LBP
ML	Sp	LW	LR	LBP
ML	Mp	SW	SR	MP
ML	Mp	SW	MR	MP
ML	Mp	SW	LR	MP
ML	Mp	MW	SR	MP
ML	Mp	MW	MR	MP
ML	Mp	MW	LR	LBP
ML	Mp	LW	SR	MP
ML	Mp	LW	MR	MP
ML	Mp	LW	LR	LBP
ML	Lp	SW	SR	SP
ML	Lp	SW	MR	SP
ML	Lp	SW	LR	LSP
ML	Lp	MW	SR	SP
ML	Lp	MW	MR	SP
ML	Lp	MW	LR	LSP

L	ρ	W	R	P
LL	Sp	SW	SR	LSP
LL	Sp	SW	MR	LSP
LL	Sp	SW	LR	LSP
LL	Sp	MW	SR	LSP
LL	Sp	MW	MR	LSP
LL	Sp	MW	LR	MP
LL	Sp	LW	SR	LSP
LL	Sp	LW	MR	SP
LL	Sp	LW	LR	LSP
LL	Mp	SW	SR	SP
LL	Mp	SW	MR	SP
LL	Mp	SW	LR	SP
LL	Mp	MW	SR	SP
LL	Mp	MW	MR	SP
LL	Mp	MW	LR	LSP
LL	Mp	LW	SR	SP
LL	Mp	LW	MR	LSP
LL	Mp	LW	LR	LSP
LL	Lp	SW	SR	SP
LL	Lp	SW	MR	SP
LL	Lp	SW	LR	SP
LL	Lp	MW	SR	SP
LL	Lp	MW	MR	SP
LL	Lp	MW	LR	SP

The consequent-oriented fuzzy inference mechanism is used to compute the output membership function for pedestrian path selection. Each fuzzy rule contributes to the final decision based on the weighted influence of four input parameters, namely path length, pedestrian density, road width, and open space. The parameters are assigned corresponding weights to reflect their relative importance in the decision-making process. Among all the parameters, path length is assigned the largest weight, indicating that it has the strongest influence on the pedestrian's path choice. The weighted contributions of all activated fuzzy rules are normalized and then aggregated to obtain the final fuzzy output representing the probability distribution of path selection. The defuzzification process is used to convert the fuzzy output into a crisp numeric probability. The centroid method is applied to obtain the final probability values corresponding to the left, center, and right paths. These probability values lie within the range of zero to one and represent the likelihood of selecting each possible path. The final movement direction of the pedestrian is determined using the maximum probability rule. The pedestrian selects the path that has the highest probability value among the left, center, and right options. This rule ensures a clear and deterministic decision for the pedestrian's future movement based on fuzzy inference.

Predicted vs actual trajectory

Predicted verses actual trajectory illustrates the performance of the proposed trajectory prediction model by visually comparing the ground-truth pedestrian paths with the future paths estimated by the algorithm. The actual trajectory represents the true motion of pedestrians extracted from the tracking data, while the predicted trajectory represents the future positions computed using the implemented prediction model. For each pedestrian, the past motion history is first observed, and based on this information, the model estimates the future path over a selected prediction horizon. Both trajectories are overlaid on the same video frame using different visual markers to clearly show the deviation between the predicted and actual motion. The difference between the predicted and actual trajectories provides a direct measure of the accuracy of the prediction model. A smaller deviation indicates better prediction performance, while larger deviations highlight regions of uncertainty or sudden changes in pedestrian behavior. This visual comparison helps in qualitatively validating the effectiveness of the proposed method in capturing real pedestrian movement patterns. The video also demonstrates the temporal consistency of the prediction, showing how closely the estimated trajectories follow the real-time pedestrian motion across consecutive frames.

3.4.3 Implementation of pedestrian Trajectory

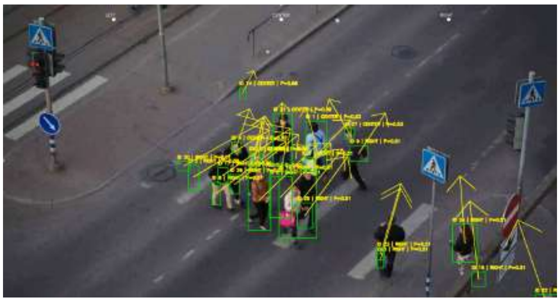

Figure 3.15: Simulation snap of direction determination

Fig 3.15 Here pedestrians direction is being shown that what the next path it will take either it went toward left , Right , or being at center.this is one of the starting path to understand the obstacle behavior and according to the obstacle make the movement.

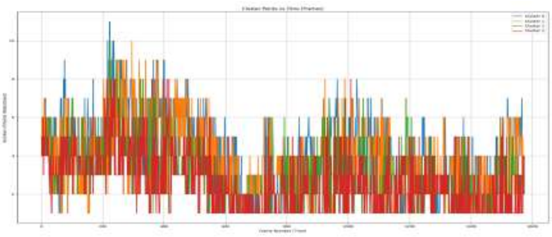

Figure 3.16: Simulation snap of cluster point vs time

Fig 3.16 Here few cluster point per frame plotting is shown that plotting helps to understand the behavior of clustering point and deping on clustering point , around that clustering point cluster is formed.

Figure 3.17: Simulation snap of predicted vs actual trajectory

Fig 3.17 Here the visualization is of predicted vs actual trajectory of pedestrians what was predicted path of pedestrian and what path they took in actually the red is showing the prediction of trajectory that should be formed but in actually yellow is formed. Cohesion Output

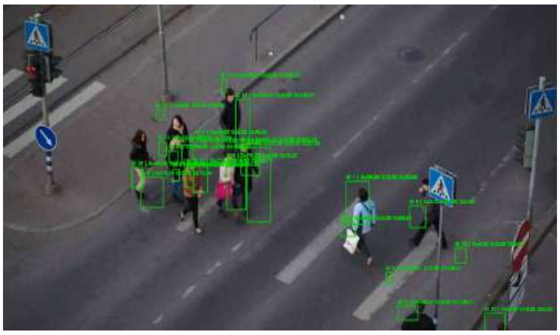

Figure 3.18: Simulation snap of all cohesion indication

Fig 3.18 Here the value of all cohesion Cp is proximity cohesion, Ci is interaction cohesion, Cw is walking speed cohesion, Cs is group size cohesion, Ct is Total cohesion. Framed around the figure is visualize that can be visible around the green round box

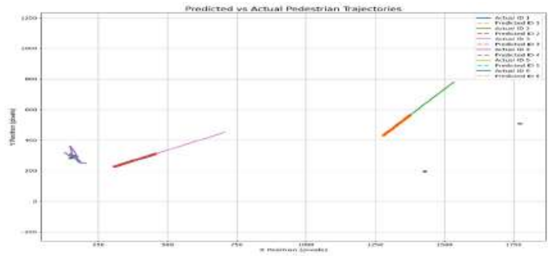

Figure 3.19: Simulation snap Predicted vs Actual Trajectory plotting

Fig 3.19 Here few pedestrians Predicted vs actual Trajectory plotting shown this trajectory shows the difference , how much deviation the plotting help to visualize.

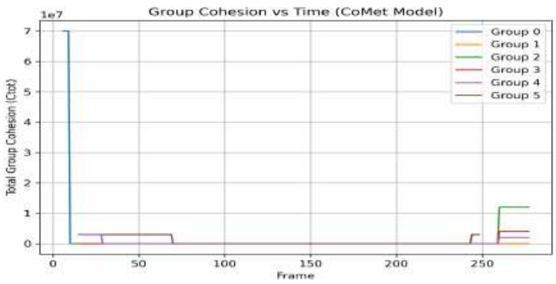

Figure 3.20: Group Cohesion plotting

Fig 3.20 Here few groups their total group cohesion per frame is shown , this help to find out group cohesion

3.5 Pedestrian age determination and attractive and repulsive cohesion analysis

Multiple cohesion components are computed and visualized to analyze pedestrian group behavior. The proximity-based cohesion evaluates how closely pedestrians remain to one another during movement, where higher values indicate stronger spatial grouping. The angle-based head rotation cohesion captures the interaction tendency among pedestrians by analyzing face orientation and relative viewing direction; higher values represent stronger social interaction within the group. The group size cohesion reflects the stability of the detected pedestrian group, where consistent group membership leads to higher cohesion values. The walking speed cohesion measures similarity in pedestrian velocities, and higher values indicate synchronized group motion. All these individual cohesion values are computed frame by frame and then combined to obtain the total cohesion of the pedestrian group. The final total cohesion represents the overall strength of group formation and coordinated movement observed in the scene. Higher total cohesion values indicate strong collective behavior, while lower values reflect weak or dispersing groups. The obtained values are displayed directly on the video frames to provide real-time visual interpretation of pedestrian group dynamics.

3.5.1 Attractive and repulsive cohesion analysis

Equations (3.56), (3.57), (3.58), (3.59), and (3.60) define the attractive and repulsive group force components (3.62), (3.63), and (3.65) define the attractive, repulsive, and overall cohesion measures, respectively [18].

Full Group Force and Cohesion Formulation

The total force acting on pedestrian i at time t is given by: [18]

$$\mathbf{F}_i(t) = \mathbf{F}_i^0 + \sum_j \mathbf{F}_{ij} + \sum_B \mathbf{F}_{iB} + \mathbf{F}_{iA} + \mathbf{F}_i^{group} \tag{3.56}$$

where $\mathbf{F}_i^{group}$ is the social group force visualized in the overlays. The group force is composed of attractive and repulsive components: [18]

$$\mathbf{F}_i^{group} = \mathbf{F}_i^{ag} + \mathbf{F}_i^{rg} \tag{3.57}$$

where $\mathbf{F}_i^{ag}$ is the attractive group force and $\mathbf{F}_i^{rg}$ is the repulsive group force.

Attractive Group Force

The attractive group force is defined as: [18]

$$\mathbf{F}_i^{ag} = A_{grp1} \cdot \mathbf{G}_i \tag{3.58}$$

where A_{grp1} is the calibrated group attractive strength. The unit direction vector from pedestrian i to the group centroid is: [18]

$$\mathbf{G}_i(t) = \frac{\mathbf{p}_{g,centroid}(t) - \mathbf{p}_i(t)}{\left\|\mathbf{p}_{g,centroid}(t) - \mathbf{p}_i(t)\right\|} \tag{3.59}$$

Repulsive Group Force

The repulsive group force is defined as:

$$\mathbf{F}_i^{rg} = \sum_k \left(S \cdot A_{grp2} \cdot \mathbf{H}_{ik}\right) \tag{3.60}$$

where A_{grp2} is the calibrated group repulsive strength, and $\mathbf{H}_{ik}$ is the unit vector from pedestrian i to group member k. The switching factor S is defined as:

$$S = \begin{cases} 1, & \text{if pedestrians } i \text{ and } k \text{ overlap} \\ 0, & \text{otherwise} \end{cases} \tag{3.61}$$

Group Cohesion Measures

The attractive cohesion is defined as:

$$C_a(i,t) = \left\|\mathbf{F}_i^{ag}(t)\right\| \tag{3.62}$$

The repulsive cohesion is defined as:

$$C_r(i,t) = \left\|\mathbf{F}_i^{rg}(t)\right\| \tag{3.63}$$

The total group cohesion or total group force is defined as: [18]

$$C_t(i,t) = \left\| \mathbf{F}_i^{ag}(t) + \mathbf{F}_i^{rg}(t) \right\| \tag{3.64}$$

An alternative representation of total cohesion is:

$$C_t = C_a + C_r \tag{3.65}$$

The group force and cohesion model represented by Equations (3.56) to (3.65) is derived from the Social Force Model and pedestrian group behavior theories, where attractive forces maintain group integrity while repulsive forces preserve interpersonal spacing and avoid overlap among group members. Similar formulations have been widely adopted for modeling pedestrian group cohesion, social interactions, and socially compliant navigation in dynamic crowd environments [7, 8, 18, 21]. The above formulation models the influence of social group behavior on individual pedestrian motion. The total force acting on each pedestrian is decomposed into personal driving force, interaction forces with other pedestrians and boundaries, and an additional group-based social force. The group force is further divided into attractive and repulsive components, which together regulate the cohesion and spacing within a pedestrian group. The attractive component pulls individuals toward the group centroid to preserve group integrity, while the repulsive component prevents excessive crowding by generating separation when pedestrians overlap. Using these forces, the attractive cohesion, repulsive cohesion, and total group cohesion are computed at each time instant. These cohesion values are visualized in the video overlays and are used to quantitatively represent the strength of group formation and coordinated movement among pedestrians in the scene.

3.5.2 Age determination modeling

Age Determination and Incorporation into Motion Model

Equations The age-dependent driving force model is described by Equations (3.66) and (3.67) (3.68), (3.69),(3.70), (3.71), (3.72) [18]. **Age Groups visual observation of video**

- **Child:** age < 16

- **Adult:** $17 \leq \text{age} \leq 60$

- **Elderly:** $\text{age} > 60$

Incorporation of Age into the Driving Force:

$$\mathbf{F}_i^0 = \frac{v_i^0(t)\mathbf{e}_i^0(t) - \mathbf{v}_i(t)}{a_{c,o,a}\,\tau_i} \tag{3.66}$$

Age Factor Definition:

$$a_{c,o,a} = \frac{\tau_{c/o/a}}{\tau_i} \tag{3.67}$$

where τ_c, τ_a, and τ_o are the age-specific relaxation times for children, adults, and elderly pedestrians, respectively. **Age Factors Used:**

$$a_c = 1.51, \quad a_a = 0.76, \quad a_o = 1.42 \tag{3.68}$$

With Base Relaxation Time:

$$\tau_i \approx 0.95 \text{ s} \tag{3.69}$$

$$\tau_c \approx 1.51 \times 0.95 \approx 1.44 \text{ s} \tag{3.70}$$

$$\tau_a \approx 0.76 \times 0.95 \approx 0.72 \text{ s} \tag{3.71}$$

$$\tau_o \approx 1.42 \times 0.95 \approx 1.35 \text{ s} \tag{3.72}$$

The age-dependent driving force model represented by Equations (3.67) to (3.66) incorporates pedestrian age characteristics into the Social Force Model by modifying the relaxation time according to age-specific behavioral responses. Similar age-based pedestrian behavior and social force formulations have been employed to model differences in walking speed, reaction time, and movement dynamics among children, adults, and elderly pedestrians [7, 10, 16]. Interpretation: Adults have the smallest age factor and therefore exhibit the highest acceleration in response to the driving force. Children and elderly pedestrians have larger age factors, resulting in slower acceleration and more gradual changes in motion. Explanation: The age of each pedestrian is estimated from visual observation of the video and classified into child, adult, or elderly categories. This age classification directly influences the pedestrian's acceleration behavior through the age-dependent driving force. By modifying the relaxation time using age-specific scaling factors, realistic motion differences among children, adults, and elderly pedestrians are introduced into the model. As a result, adults respond faster to desired velocity changes, while children and elderly pedestrians move more cautiously with slower acceleration. This improves the realism of pedestrian motion modeling in the simulation. Robots were tested in a simulated environment using Gazebo under various crowd densities and dynamic obstacle conditions. The navigation system demonstrated effective performance, with the robots successfully avoiding collisions and exhibiting socially acceptable behaviors such as maintaining personal space and responding to group dynamics.

3.5.3 Implementation of age determine

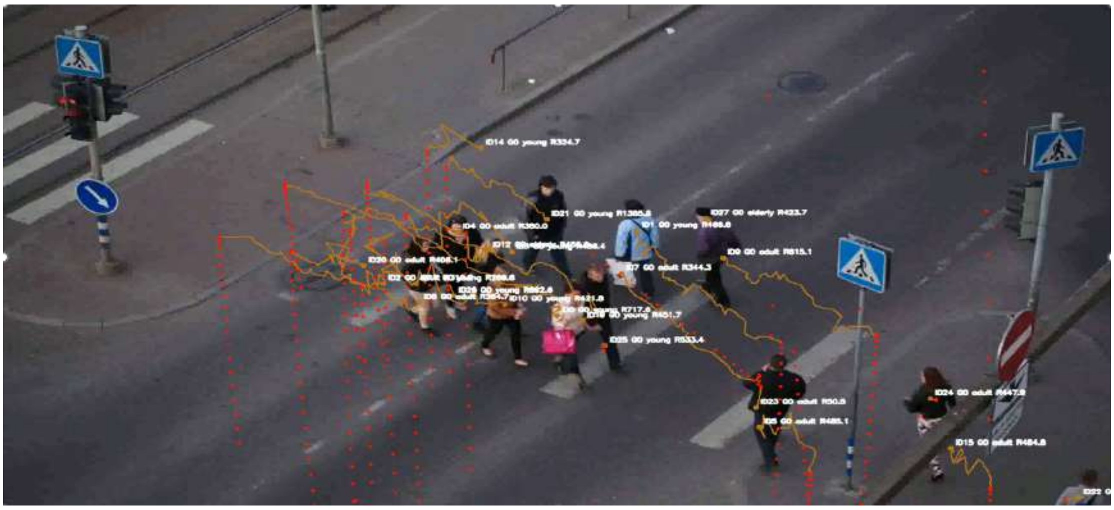

Figure 3.21: Simulation snap of age determination

Fig3.21 Here every pedestrians their age is determined either they are elderly, Adult or child based on the prediction help to find the pedestrian age determination help to make the movement in according to understand the behaviour.

3.6 Summary

This chapter covers the vast area of research from complete understanding of pedestrians lies in this area, so basically this chapter arises with first level of understanding of all the cohesion in the social environment then these cohesion of either interaction cohesion, group cohesion, proximity cohesion or any help to find the covering of overall cohesion either using any of the machine learning algorithm or to identify uses YOLO, or DBSCAN to cluster Or use Deepsort to count the number is there and further this chapter moves towards the grouping all the pedestrians based on the the combination of these cohesions, and also help to find some small factory which also the part of social behaviour to not underestimate these factors to under find limitation in this chapter also, so we take the consideration of other factors like Personal comfort distance Modeling, knowing the group interaction detection, and group interacting modeling behaviour also understanding the different pedestrians dynamic modeling that is seen to be applied using machine learning, and taking consideration of pedestrian Trajectory using either Gaussian Membership functions for direction of trajectory estimation of these pedestrians then apply their final decision rule to be followed to make trajectory based on Fuzzy Path Decision condition to be applied to use their and make the trajectory to be complete then have to be seen so many graphs give the understanding of the trajectory and shows the difference visualization in predicted vs actual trajectory that we generally cannot understand then applied apart from this here as a social factor applied the age determination and proportionate in the model to identify the person there is adult, young or old. This chapter last with understanding the pedestrian trajectory and make the trajectory to be completed.

Chapter 4

Clustering analysis

4.1 Introduction

clustering involves breaking up a data into naturally occurring subgroups where each group's members share a large number of characteristics. Clustering is a technique where no prior assumptions are made about the number of groups. The essential features that need to be dealt with while discussing a clustering technique are how many clusters to be formed, within the cluster itself, and how to measure the distance between different clusters. The validity index of clustering helps to validate of the clusters formed. Below is the algorithm

4.1.1 Bisecting K-Means algorithm

Here algorithm is [27]

1. Initialize the cluster list with single cluster containing all data points.

2. Repeat until the desired number of clusters K is obtained:

   - Select and remove one cluster from the current cluster list.
   - Perform n trial bisections of the selected cluster using K-Means with $k = 2$.
   - Compute the Sum of Squared Errors (SSE) for each trial.
   - Select the bisection with the minimum SSE.
   - Add the two resulting clusters to the cluster list.

3. Stop when the cluster list contains exactly K clusters and output the final clusters.

4.1.2 Density based spatial clustering of applications with noise algorithm

The algorithm DBSCAN (Density Based Spatial Clustering of Applications with Noise) which is designed to discover the clusters and the noise in a spatial database. Ideally, There would have to know the appropriate parameters Eps and MinPts of each cluster and at least one point from the respective cluster. Then, it could retrieve all points that are density from the given point using the correct parameters. But there is no easy way to get this information in advance for all clusters of the database. However, there is a simple and effective heuristic to determine the parameters Eps and MinPts of the ”thinnest”, i.e. least dense, cluster in the database. Therefore, DBSCAN uses global values for Eps and MinPts, i.e. the same values for all clusters.The density parameters of the ”thinnest” cluster are good candidates for these global parameter values specifying the lowest density which is not considered to be noise. Algorithm with [27]

Algorithm

1. Select the input parameters: Neighborhood radius (ε) and minimum number of points ($MinPts$).
2. Mark all data points as *UNCLASSIFIED*.
3. Randomly select an unclassified point and determine its ε-neighborhood.
4. If number of neighboring points is less than $MinPts$, mark the point as *NOISE*; otherwise, create a new cluster and assign a unique Cluster ID.
5. Add all points in the ε-neighborhood to the current cluster.
6. For each point in the cluster, compute its ε-neighborhood. If it contains at least $MinPts$ points, add all density-reachable neighbors to the cluster.
7. Continue expanding the cluster until no additional points can be added.

8. Repeat Steps 3-7 until all points are assigned to a cluster or labeled as noise.

9. Output final set of clusters and noise points.

4.1.3 Automated robust clustering algorithm

The proposed automated clustering algorithm begins with the entire dataset $A(n \times p)$ as input and generates disjoint clusters iteratively.Algorithm below [28]

1. Compute the robust location estimate m and dispersion matrix S of A. Calculate the Mahalanobis distance $A_s(m, P_j)$ for each point P_j. If $A_s(m, P_j) < R(C_i)$, assign P_j to cluster C_i.

2. Compute the robust location estimate m_i and dispersion matrix S_i of cluster C_i. Recalculate the Mahalanobis distance for all points in C_i and update the cluster until its final shape $\bar{C}_i$ is obtained.

3. Repeat the above steps for the remaining data points until all observations are assigned to disjoint clusters.

Algorithm

1. Initialize data $= X$, compute $n = \text{rows}(X)$ and $m = \text{cols}(X)$.

2. If $n \leq 1$, terminate; otherwise compute and print d and d_{rest}.

3. If $n_1 > 2m + 1$, set $d = d_{\text{rest}}$ and update $n_1 = \text{rows}(d_{\text{rest}})$.

4. If $n_1 \leq 1$, return to Step 2; otherwise continue.

5. Compute d_1 and d_2 from d, where $n_1 = \text{rows}(d)$ and $n_3 = \text{rows}(d_1)$.

6. If $n_1 = n_3$, set $d = d_1$; otherwise set data $= d_2$ and update $n = \text{rows}(d)$.

7. If the stopping criterion is satisfied, print d_2 and terminate.

8. Otherwise, return to Step 2 and repeat until convergence.

4.2 Implementation of Different Clustering

Figure 4.1: Simulation snap of bisecting K- means clustering

Fig4.1 Here bisecting K- means clustering is applied where k pedestrian around the cluster center is available and this clustering help to find the clustering belonging to same group help the classification of 1 group to another.

Figure 4.2: Simulation snap of DBSCAN clustering

Fig4.2 Here DBSCAN clustering is performed and main group is consider group and other are act as noise , this clustering provide better solution helping to remove the noise and make perfect clustering

Figure 4.3: Simulation snap of automated robust clustering

Fig4.3Here Robust automated clustering method is applied to group the number of pedestrian according to the size of people , this almost consider 3dimensional view that can be viualize in figure.

4.3 Summary

This chapter touch the area of clustering to make the group of people in 1 cluster se ction, if similar group is align in some vast number of circle to understand further for future approaching , how to align in these pedestrian groups, so in this chapter have to cover the understanding the groups and visualize through 3d clustering by taking one as cluster center and form the group . This group helps to visualize the different groups in single frame , so that further approaching towards the pedestrian by the robot is easier.so here firdt cover simple k- means clustering where it is to be find that the this is very simple and complete group of people is not covered the went fror other method that is DBSCAN that also shows some limitation then went for Automated clustering methods which more or less cover the problem have include all people around a cluster and form a complete cluster , helps to visualize the clusters and understand these clustering pattern . but their also find some limitation that is it wont work on 3d visualization have to be find that their is some lagging has to seen that not perfectly align as the pereson is trying to align in 1 cluster then it make some arbitary cluster that shows that it has also limitations that in the future need to be cover.

Chapter 5

Experimental setup and hardware deployment

5.1 Introduction

The human detection module was implemented using the YOLOv5 , deep learning framework. The model was deployed on a laptop camera and successfully detected pedestrians in real time with high accuracy and low latency. YOLOv5 processed each video frame and generated bounding boxes around detected persons while providing confidence scores for every detection. The system demonstrated reliable performance under varying lighting conditions and pedestrian densities. To maintain continuous observation of detected individuals across multiple video frames, the DeepSORT (Deep Simple Online Realtime Tracking) algorithm was integrated with YOLOv5. While YOLOv5 provided frame-by-frame detections, DeepSORT assigned a unique identification number (ID) to each pedestrian and maintained that ID throughout the tracking process. This combination enabled robust multi-object tracking even when pedestrians temporarily occluded each other or moved through crowded environments. The tracking system generated a continuous trajectory for each pedestrian by storing the center coordinates of the detected bounding boxes over time. These trajectories were recorded in CSV files and used to estimate pedestrian motion characteristics such as position, velocity, direction of movement, and turn angle. The generated trajectories formed the foundation for higher-level crowd analysis, including group detection, cohesion estimation, and social interaction modeling.

After successful detection and tracking using the laptop camera, the next phase of the implementation involves deploying the system on the TurtleBot3 Waffle Pi platform. The RGB camera mounted on the Waffle Pi will act as the primary sensing device for real-time pedestrian perception. The live camera feed will be processed using the same YOLOv5 and DeepSORT pipeline to identify and track surrounding pedestrians within the robot's environment. The tracked pedestrian trajectories will then be integrated with the robot navigation framework. By analyzing the movement patterns of pedestrians, the robot can estimate safe navigation corridors and make the movement. The trajectory information will be utilized to generate socially aware navigation decisions, allowing the robot to avoid collisions while maintaining smooth motion in dynamic environments.

For autonomous navigation, a goal location will be specified within the environment. The TurtleBot3 Waffle Pi will continuously compute its position relative to the goal and generate a collision-free trajectory toward the destination. During navigation, the robot will use pedestrian trajectory predictions to dynamically update its path whenever moving obstacles are detected.

Furthermore, the collected trajectory data can be used to compute pedestrian velocity, acceleration, group cohesion scores, and interaction metrics. These parameters support advanced crowd-aware navigation strategies where the robot not only avoids individual pedestrians but also preserves social group structures. As a result, the navigation system becomes more socially compliant and suitable for deployment in crowded public spaces such as shopping malls, airports, railway stations, and university campuses. The integration of YOLOv5 for detection, DeepSORT for tracking, trajectory prediction for motion estimation, and TurtleBot3 Waffle Pi for autonomous navigation creates a complete perception-to-navigation pipeline capable of real-time operation in dynamic human-populated environments. This framework provides a strong foundation for future extensions involving social force models, reinforcement learning-based navigation, group-aware path planning, and human-robot interaction studies.

5.2 Experimental results of hardware

Figure 5.1: Experimental Result of Waffle pi trajectory to reach to the goal

Fig5.1 Here a waffle pi Robot (Hardware) is there to demonstrate the presence of pedestrians perform cohesion and clustering algorithmn through laptop to detect the person, so detection is determined and show in bounding purple box of all the pedestrian by the camera attach and then waffle pi robot make the trajectory to reach to the goal.

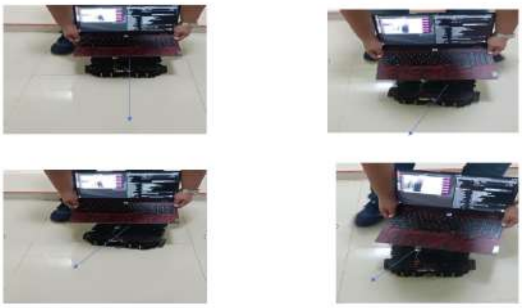

Figure 5.2: Experimental Result of Hardware implement and visual of trajectory directions

Fig 5.2 Here a waffle pi Robot (Hardware) is there to demonstrate the presence of pedestrians perform cohesion and clustering algorithm through laptop to detect the person, so detection is determined and show in bounding purple box of all the pedestrian by the camera attach and then waffle pi robot make the trajectory to reach to the goal and showing 4 position that is used to determine the trajectory to be shown.

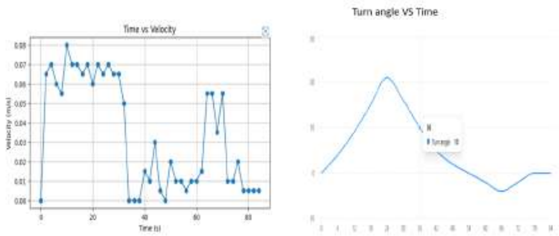

Figure 5.3: Experimental Result of velocity and turn angle plots

Fig 5.3 plotting of the movement of robot so that this velocity is find with the time and also the change in turn angle vs time graph is visualize for the turning of robot after finding the obstacle , so we obtain these trajectory by waffle pi making the trajectory in this time frame and fluctuation is also visualize and other graph shown the turn angle of the waffle pi with the considering the angle.

5.3 Summary

This last fifth chapter covers the area of Hardware implementation that we discussed in the last three chapters. This chapter tries to focus on the area where we can implement the simulation from the past three chapters in hardware. This chapter remains the toughest of all because implementing in hardware is really challenging. Here, I try to cover the use of machine learning and one of the algorithms to visualize it through the camera to identify pedestrians, count the number of people present, and analyze their clustering behavior. We will also examine their trajectories, showing the dynamic behavior of the robot, and then create the trajectory of the robot accordingly with this turtle wafflepi is used to take APF and make the navigation of the trajectory to implementand make a collision free trajectory to reach to the goal.

Chapter 6

Conclusion

This project presented a complete framework for socially intelligent navigation of mobile robots in dynamic, human-populated environments. By combining real-time perception, behavior-based planning, and multi-robot coordination strategies, the system demonstrates safe, efficient, and human-aware navigation behavior.

Pedestrian Modeling:The implemented pedestrian modeling pipeline effectively integrates detection, tracking, velocity estimation, trajectory prediction, group detection, and cohesion analysis to capture both individual motion and social behavior in dynamic environments.

The conclusion is that this robot work finely with this way but need so many midification to work well all the research gap that is to be find the need to look on that area.

Bibliography

[1] Z. Xie and P. Dames, “DRL-VO: Learning to Navigate Through Crowded Dynamic Scenes Using Velocity Obstacles,” *IEEE Transactions on Robotics*, vol. 39, no. 4, Aug. 2023.

[2] R. K. Megalingam, A. R., D. H. T. A. Babu, G. Sriram, and V. S. Y. Avvari, “Implementation of a Person Following Robot in ROS-Gazebo Platform,” in *Proc. International Conference for Advancement in Technology (ICONAT)*, 2022.

[3] S. Bandini, A. Gorrini, L. Manenti, and G. Vizzari, “Crowd and Pedestrian Dynamics: Empirical Investigation and Simulation,” in *Proc. Measuring Behavior*, 2012.

[4] Y. F. Chen, M. Everett, M. Liu, and J. P. How, “Socially Aware Motion Planning with Deep Reinforcement Learning,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2017.

[5] L. Tai, J. Zhang, M. Liu, and W. Burgard, “Socially Compliant Navigation Through Raw Depth Inputs With Generative Adversarial Imitation Learning,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2018.

[6] A. J. Sathyamoorthy, U. Patel, T. Guan, and D. Manocha, “Frozone: Freezing-Free, Pedestrian-Friendly Navigation in Human Crowds,” *IEEE Robotics and Automation Letters*, .

[7] D. Helbing and P. Molnar, “Social Force Model for Pedestrian Dynamics,” *Physical Review E*, 1995.

[8] M. Moussaïd, N. Perozo, S. Garnier, D. Helbing, and G. Theraulaz, "Walking Behaviour of Pedestrian Social Groups and Its Impact on Crowd Dynamics," *PLoS ONE*, vol. 5, 2010.

[9] S. Pellegrini, A. Ess, K. Schindler, and L. Van Gool, "You'll Never Walk Alone: Modeling Social Behavior for Multi-Target Tracking," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2009.

[10] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple Online and Realtime Tracking," in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, 2016.

[11] N. Wojke, A. Bewley, and D. Paulus, "Simple Online and Realtime Tracking With a Deep Association Metric," in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, 2017.

[12] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human Trajectory Prediction in Crowded Spaces," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016.

[13] W. Wang, "Research on Modeling and Simulation of Pedestrian Street Motion Based on Fuzzy Inference and Path Selection," in *Proc. International Conference on Computer Simulation and Modeling Information Security (CSMIS)*, 2023.

[14] S. M. P. Siddharth and V. Perumal, "Development of the Social Force Model Considering Pedestrian Characteristics and Behavior," 2023.

[15] J. A. Placed, J. Strader, V. Indelman, J. A. Castellanos, H. Carrillo, N. Atanasov, and L. Carlone, "A Survey on Active Simultaneous Localization and Mapping: State of the Art and New Frontiers," *IEEE Transactions on Robotics*, vol. 39, 2023.

[16] X. Chen, M. Treiber, V. Kanagaraj, and H. Li, "Social Force Models for Pedestrian Traffic – State of the Art," 2023.

[17] M. Zong, Y. Chang, Y. Dang, and K. Wang, "Pedestrian Trajectory Prediction in Crowded Environments Using Social Attention Graph Neural Networks," 2023.

[18] G. Vizzari, L. Manenti, and L. Crociani, "Adaptive Pedestrian Behaviour for the Preservation of Group Cohesion," *Complex Adaptive Systems Modeling*, vol. 1, no. 7, 2013.

[19] F. Fang, X. Wang, Z. Li, K. Qian, and B. Zhou, "A Unified Framework for Pedestrian Trajectory Prediction and Social-Friendly Navigation," *IEEE Transactions on Industrial Electronics*, vol. 71, no. 9, 2024.

[20] V. B. Hoang, V. H. Nguyen, T. D. Ngo, and X.-T. Truong, "Socially Aware Robot Navigation Framework: Where and How to Approach People in Dynamic Social Environments," *International Journal of Social Robotics*, 2023.

[21] A. J. Sathyamoorthy, U. Patel, M. Paul, N. K. S. Kumar, Y. Savle, and D. Manocha, "CoMet: Modeling Group Cohesion for Socially Compliant Robot Navigation in Crowded Scenes," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, 2022.

[22] X.-T. Truong and T. D. Ngo, "Toward Socially Aware Robot Navigation in Dynamic and Crowded Environments: A Proactive Social Motion Model," *IEEE Transactions on Automation Science and Engineering*, vol. 14, 2017.

[23] N. Hirose, D. Shah, A. Sridhar, and S. Levine, "SACSoN: Scalable Autonomous Control for Social Navigation," arXiv preprint, 2024.

[24] X. Lu, A. Faragasso, Y. Wang, H. Asama, and A. Yamashita, "Group-Aware Robot Navigation in Crowds Using Spatio-Temporal Graph Attention Network With Deep Reinforcement Learning," *IEEE Robotics and Automation Letters*, 2025.

[25] Jordan Fraser, "Understanding DeepSORT: Real-Time Object Tracking Explained," Medium, Available: `https://jordanfraser.medium.com`. Accessed: Jun. 2026.

[26] M. Steinbach, G. Karypis, and V. Kumar, "A Comparison of Document Clustering Techniques," in Proc. KDD Workshop on Text Mining, Boston, MA, USA, 2000, pp. 109–110.

[27] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, “A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise,” in *Proc. 2nd Int. Conf. Knowledge Discovery and Data Mining (KDD)*, Portland, OR, USA, 1996, pp. 226–231.

[28] R. N. Dave and R. Krishnapuram, “Robust Clustering Methods: A Unified View,” IEEE Transactions on Fuzzy Systems, vol. 5, no. 2, pp. 270–293, May 1997.